How well does task EEG separate major depression from healthy controls in participants never seen by the model?

A leakage-free, pre-specified evaluation on the MODMA 128-channel dot-probe dataset

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Abstract

Background. Machine-learning studies often report near-perfect accuracy for detecting major depressive disorder (MDD) from EEG. Many of them, however, split trials rather than participants between training and test sets, and few compare against non-EEG baselines.
Methods. We re-analysed the 128-channel dot-probe ERP recordings of the MODMA dataset (24 MDD, 29 HC) from the raw files with a pre-specified, version-controlled plan. Preprocessing used PREP bad-channel detection and ICA with ICLabel, and participants were excluded by fixed quality criteria. We evaluated 18 model configurations:

- a demographic baseline;
- linear, kernel and tree models on 288 handcrafted features (spectral, frontal alpha asymmetry, ERP, wPLI connectivity, complexity);
- Riemannian models;
- three deep networks on full epochs.

All models used nested cross-validation over participants with identical folds. Results carry participant-level bootstrap confidence intervals, and all but the deep networks carry permutation tests of the full pipeline.
Results. In 45 participants, the pre-specified primary model (L2 logistic regression on all features) reached AUC 0.53 (95% CI 0.36–0.70, $p = 0.456$). No EEG model exceeded a baseline using only age, sex and education (AUC 0.78). The best secondary model (AUC 0.72) did not survive multiplicity correction, and nested model selection gave AUC 0.63. Evaluated with an invalid trial-wise split, the same tangent-space model achieved apparently perfect participant-level classification (AUC 1.00), versus 0.50 when participants were held out. The null result held when all 53 participants were included and when the analysis was restricted to a period in which both groups were recorded.
Conclusions. Evaluated on unseen participants, task EEG in this dataset carries no detectable diagnostic signal for MDD beyond demographics. Trial-wise evaluation can manufacture high accuracy. Participant-wise evaluation, pre-specification and demographic baselines should be standard.

1 Introduction

Electroencephalography (EEG) is cheap, portable and widely available, which makes it an attractive candidate for objective markers of major depressive disorder (MDD). Many machine-learning studies report high accuracies for separating patients with MDD from healthy controls (HC) using EEG. However, a large share of this literature evaluates models by splitting *trials* or *segments*, rather than *participants*, between training and test sets.

When the same person contributes data to both sets, a classifier can succeed by recognising the individual (their head geometry, impedances, idiosyncratic rhythms or recording session) instead of the disorder. The resulting estimates do not approximate the intended use-case, which is diagnosing a new person [1]. This form of leakage is a documented driver of the reproducibility

crisis in ML-based science [2]. It has recently been shown to inflate performance substantially in translational EEG studies [3].

Small samples aggravate the problem. With a few dozen participants, cross-validated accuracy has error bars of ± 10 percentage points or more [4], and theoretical chance levels are not attained in practice [5]. Finally, demographic or acquisition variables that differ between groups (sex, education, recording period) can masquerade as disorder-related signal.

The Multi-modal Open Dataset for Mental-disorder Analysis (MODMA) [6] provides 128-channel EEG recorded during an emotional-face dot-probe task [7] from patients with MDD and matched controls. It is a natural testbed for these questions. In this paper we ask a single question: *how well can task EEG classify MDD versus HC in participants that the model has never seen?*

Contributions.

1. A complete pipeline from the raw EGI files, with exact ID matching against the clinical spreadsheet, event decoding by channel name, per-participant preprocessing (PREP bad-channel detection, ICA with ICLabel) and pre-specified participant exclusion criteria.

2. An analysis plan committed to version control before any result existed. It fixes the primary model, the primary outcome and all thresholds, and every deviation is reported.

3. A strictly subject-wise evaluation of 18 model configurations across five families: a demographic baseline, handcrafted biomarkers (spectral, frontal alpha asymmetry, ERP, connectivity, complexity), Riemannian geometry and three deep networks.

    - Every model uses nested cross-validation and identical subject partitions.
    - Every result carries a participant-level bootstrap confidence interval; all models except the deep networks also carry a label-permutation test of the full nested pipeline.

4. Confound analyses: can EEG predict sex, age or recording period, and does performance survive restriction to a period in which both groups were recorded?

5. A direct measurement of how much trial-wise splitting inflates performance on the same data.

2 Related work

EEG markers of depression. Reviews of EEG markers of MDD have catalogued many candidates. These include quantitative band power, frontal alpha asymmetry (FAA), connectivity, vigilance, sleep EEG and event-related potentials. Effects are generally small and heterogeneous across studies, and the reviews repeatedly call for standardised procedures [8–10].

The most studied single marker is FAA [11]. A meta-analysis concluded that it has no diagnostic value for MDD [12].

Emotional ERPs and the dot-probe task. The dot-probe paradigm measures attentional bias towards emotional stimuli [7]. ERP studies of emotional faces typically examine three components:

- the face-sensitive N170 [13];
- the P300, indexing attentional allocation [14];
- the late positive potential (LPP), indexing sustained motivated attention [15].

Table 1: Sample characteristics (all 53 recordings). Mean ± SD. †Descriptive only; never used as predictors.

Variable	MDD ($n = 24$)	HC ($n = 29$)	Test	p
Age (years)	30.9 ± 10.4	31.4 ± 9.1	Mann–Whitney	0.82
Education (years)	13.2 ± 3.8	16.0 ± 2.3	Mann–Whitney	0.0048
PHQ-9†	18.3 ± 3.5	2.7 ± 1.8	Mann–Whitney	<0.001
GAD-7†	13.4 ± 4.9	2.1 ± 2.1	Mann–Whitney	<0.001
PSQI†	12.5 ± 4.2	4.1 ± 2.2	Mann–Whitney	<0.001
Sex (F/M)	11/13	9/20	Fisher exact	0.39

The Lanzhou group that later released MODMA used a dot-probe task with emotional–neutral face pairs in 17 patients and 17 controls. They reported larger P300 amplitudes when the probe replaced a sad face, interpreted as difficulty disengaging from negative information, and explored ERP components as classification features [16]. The 128-channel ERP subset of MODMA (24 MDD, 29 HC) uses fear-, sad- and happy-neutral face pairs [6].

Machine learning on MODMA and on depression EEG in general. Studies applying machine learning to depression EEG often report accuracies above 90%. For example, a CNN–LSTM applied to EEG segments reached 97.7–99.1% [17].

Such figures depend critically on how data are split. When segments of the same participant appear in both training and test sets, a model can identify participants instead of the disorder. Segment-wise evaluation has been shown to overestimate performance in translational EEG [3], consistent with a broader literature on record-wise splitting and leakage [1, 2].

Fewer MODMA studies use subject-independent protocols. A graph-network approach on the 128-channel *resting* EEG, with leave-one-out validation, reported around 81% accuracy for multi-class severity recognition [18].

Published MODMA studies differ in signal type, preprocessing and validation, and often under-report whether folds are formed over participants. We therefore do not attempt a quantitative comparison. Instead, Section 4.3 measures directly, on the same data and pipeline, how much trial-wise splitting changes the estimate.

Small samples. With $n \approx 50$, cross-validated estimates have wide confidence intervals [4]. Theoretical chance levels are also unreliable, so permutation-based significance testing is required [5]. Choosing the best of several models on the same cross-validation inflates its estimate [19]. These three facts motivate our pre-specified primary model, our participant-level bootstrap intervals, our full-pipeline permutation tests and our nested model-selection estimate.

3 Methods

3.1 Participants and data

We used the 128-channel event-related potential (ERP) recordings of MODMA (Lanzhou, 2014–2016) [6]. The release contains 53 recordings, one per participant, sampled at 250 Hz. They were acquired with a HydroCel Geodesic Sensor Net (EGI) using 128 scalp electrodes plus the vertex reference (E129/Cz).

Diagnostic group (24 MDD, 29 HC) was taken *exclusively* from the official subject-information spreadsheet. Recording files were matched to it by exact identifier, without normalisation: all 53 identifiers matched one-to-one.

Table 1 summarises the sample. The groups did not differ in age ($p = 0.823$) or sex ($p = 0.394$), but MDD participants had fewer years of education ($p = 0.005$). Questionnaire scores (PHQ-9, GAD-7, CTQ-SF, LES, SSRS, PSQI) were used only to describe the sample and never as predictors, because they measure, or are closely tied to, the target.

Task and events. Each trial of the dot-probe task [7] consisted of the following sequence:

1. a fixation cross (464–652 ms);
2. a pair of faces, one emotional and one neutral (516 ms);
3. an inter-stimulus interval (116–316 ms);
4. a dot probe (168 ms), followed by the participant's response.

Trials came in three blocks of 160 (fear, happy, sad), with block order varying across participants.

We decoded events from the individual EGI event channels by name (`fcue`, `hcue`, `scue`). The combined stimulus channel produced by the reader carries the same code for every event in these files and is therefore unusable. Every recording contained the complete 480 trials. Epochs were locked to face-pair onset, because the probe appears at least 632 ms after it, so the window up to 600 ms is free of probe-evoked activity.

3.2 Pre-specified analysis plan

Before any epoch, feature or model output existed, we committed an analysis plan to version control. It fixed the preprocessing, the participant exclusion criteria, the feature definitions, the evaluation protocol, the primary model and the primary outcome.

Five deviations are listed in Appendix A: four were decided before any model was run, and the fifth (a computational shortcut for the Riemannian models) before any of their results existed. The only one affecting the data was replacing an ad-hoc bad-channel rule with the PREP implementation (below), after the original rule produced implausible flags next to the reference electrode.

3.3 Preprocessing

All steps were performed separately for each participant with MNE-Python [20], so no information is shared across participants:

1. Bad channels were detected with the PREP pipeline (PyPREP `NoisyChannels`, without RANSAC) [21]. The criteria were NaN/flat signal, deviation, high-frequency noise, low correlation, low SNR, dropout and abnormal PSD. The vertex reference was exempt, being flat by construction.
2. Independent component analysis (Picard, extended-infomax objective [22], 30 components) was fitted on a copy filtered at 1–100 Hz, notched at 50 Hz (the mains frequency in China) and average-referenced, with bad channels excluded.
3. Components were labelled with ICLabel [23, 24]. A component was removed when its most probable class was eye, muscle, heart, line noise or channel noise *and* that class had probability ≥ 0.80.
4. The analysis copy was band-pass filtered at 0.5–40 Hz (zero-phase FIR) and notched at 50 Hz. It was then average-referenced, the ICA cleaning was applied, bad channels were interpolated (spherical splines), and it was re-referenced to the average.
5. Epochs spanned -200 to 600 ms, with a -200 to 0 ms baseline. An epoch was rejected if any channel exceeded $\pm 100\,\mu$V.
6. For spectral, connectivity and complexity features, the task period of the same cleaned recording was resampled to 125 Hz and cut into 2-s segments. The same amplitude rule was applied to the segments.

Participant exclusion (pre-specified). A participant was excluded if any of these held:

- more than 20% of channels were bad;
- fewer than 240 of 480 trials (50%) survived rejection;
- fewer than 40 trials survived in any condition.

3.4 Features

All features were computed per participant, without labels. They are therefore not a source of leakage; everything that *learns* from data happens inside the cross-validation loop.

Regions. We defined ten regions: frontal, central, temporal, parietal and occipital, each split into left and right (Fig. 8 in the appendix). Each HydroCel electrode was assigned to the region of its nearest 10–10 anchor position on the fitted head sphere. Midline electrodes and the lower face/neck ring (more than 115° from the vertex) were not used for regional features.

Feature families.

- **Spectral (100 features).** Welch power in the delta (1–4 Hz), theta (4–8), alpha (8–13), beta (13–30) and low-gamma (30–40 Hz) bands, as $\log_{10}$ absolute and relative power per region.
- **Frontal alpha asymmetry (2).** $\ln \alpha_{\text{right}} - \ln \alpha_{\text{left}}$ for the HydroCel equivalents of F4/F3 (E124/E24) and F8/F7 (E122/E33) [11, 12].
- **ERP (21).** Measured per condition on the participant's average, using windows fixed a priori from the literature:
  - N170 peak amplitude and latency at left and right occipito-temporal clusters (around P7/P8), 140–200 ms [13];
  - P300 mean amplitude (250–500 ms) and peak latency at a centro-parietal cluster (around Pz) [14];
  - late positive potential (LPP), mean amplitude 400–600 ms, at the same cluster [15].
- **Connectivity (135).** Weighted phase-lag index (wPLI) [25] between the ten region-mean signals, in the theta, alpha and beta bands.
- **Complexity (30).** Per region, sample entropy [26], Higuchi fractal dimension [27] and Lempel–Ziv complexity [28], averaged over 2-s segments.

This gives 288 subject-level features. For trial-level models we kept the ten region-mean signals of each epoch (10×201 samples) and, for deep learning, the full 129×201 epochs.

3.5 Models

- **Demographic baseline.** L2 logistic regression on age, sex and education. It indicates how much of any EEG result could be explained by demographics alone.
- **Handcrafted features.** Each model is a scikit-learn [29] pipeline: median imputation, standardisation, then the classifier.
  - *Primary, fixed a priori:* L2-regularised logistic regression (LR-L2) on all 288 features.
  - *Secondary:* L1-LR, linear and RBF support-vector machines, random forest [30] and depth-limited LightGBM [31], plus LR-L2 on each feature family separately.

Picking the best of many models inflates its estimate [19]. We therefore also report a *nested model selection* estimate, in which the inner loop chooses both the model family and its hyperparameters.

- **Riemannian, trial level.** [32, 33]
  - OAS-shrunk covariance matrices [34] of the regional epoch signals, either projected to the tangent space and classified with LR, or classified with minimum distance to mean (MDM);
  - ERP-augmented covariances, whose class prototypes are estimated inside each training fold, followed by tangent space and LR with a fixed $C = 0.1$ (deviation D5).
- **Deep learning, trial level.** EEGNet [35], ShallowFBCSPNet [36] and a temporal convolutional network (TCN) [37, 38], as implemented in braindecode.
  - Training used AdamW (learning rate 10^{-3}, weight decay 10^{-3}), cosine annealing, class-weighted cross-entropy, and augmentation by random time shifts (± 40 ms), amplitude scaling and Gaussian noise.
  - Early stopping (patience 10, maximum 60 epochs) used the subject-level AUC on a validation split of the *training participants* (20%).
  - Channel-wise normalisation statistics came from the inner training participants only.

Trial-level models produce one probability per trial. These were averaged within each participant to give a single participant-level prediction.

3.6 Evaluation protocol

- **Outer loop.** Stratified 5-fold cross-validation over *participants*, repeated 10 times (3 times for deep learning, for computational reasons). All models were evaluated on *identical* participant partitions.
- **Inner loop.** Hyperparameters were selected by 5-fold participant-grouped cross-validation within the training participants, maximising AUC.
- **Leakage check.** An assertion verifies, in every outer, inner and validation split, that no participant contributes data to both sides.
- **Metrics.** All metrics are computed per participant: ROC AUC (the pre-specified primary outcome), balanced accuracy, sensitivity and specificity at a fixed 0.5 threshold (never tuned on test data), F1 and Brier score.
- **Uncertainty.** Point estimates are averaged over repetitions. 95% confidence intervals come from 2,000 stratified bootstrap resamples of *participants* applied to the out-of-fold predictions, with interval bounds averaged over repetitions.
- **Significance.** Label-permutation tests [5] permute labels *between participants* and re-run the entire nested pipeline. Fast models used 1,000 permutations; slower models used 200. We report $p = (1 + \#\{\mathrm{AUC}_{\mathrm{perm}} \geq \mathrm{AUC}_{\mathrm{obs}}\})/(1 + n_{\mathrm{perm}})$.

3.7 Confounds, sensitivity and inflation control

- **Can EEG predict confounds?** We applied the primary pipeline to three alternative targets: sex, age above the median, and recording period.
  - The recording period is relevant because group and period are partly confounded: between October 2015 and January 2016 only controls were recorded, and in March–April 2016 only patients.

  - We therefore also re-ran the primary pipeline on the participants recorded between June and August 2015, when both groups were recorded in parallel.

- **Exclusions.** Because exclusions were more frequent in MDD, we re-ran the primary model on all 53 participants.

- **Signal quality.** We compared retained trials, interpolated channels and removed ICA components between groups.

- **Inflation control (invalid by design).** To quantify the inflation caused by trial-wise splitting, we evaluated two trial-level models with both a trial-wise split and a subject-wise split, keeping everything else identical:

  - the tangent-space model;
  - a random forest on raw regional amplitudes, mimicking common practice.

3.8 Interpretation

- **Permutation importance.** For the primary model, computed on the *test* folds of the outer loop, for single features and for whole feature families permuted jointly.

- **SHAP.** SHAP values [39] of the primary model refitted on all participants (descriptive).

- **Univariate effects.** Cohen's d with Mann–Whitney tests and Benjamini–Hochberg false-discovery-rate correction [40].

- **Cluster-based permutation tests** [41] on channel-level band power (spatial clusters) and on ERPs (spatio-temporal clusters), as descriptive neurophysiology.

4 Results

4.1 Sample and signal quality

Of the 53 recordings, 45 met the pre-specified quality criteria (18 MDD, 27 HC). Exclusions affected 6 MDD and 2 HC participants (Fisher's exact $p = 0.12$). Among included participants, the median number of retained face-locked trials was 427 of 480 (range 265–471), the median number of interpolated channels was 7 (range 1–25) and the median number of removed ICA components was 7 of 30 (range 1–13). None of these quality measures differed between groups (Mann–Whitney $p = 0.36$, 0.32 and 0.92, respectively; Fig. 7). Any group difference in the features is therefore unlikely to reflect a gross difference in data quality.

4.2 Classification of MDD vs. HC

Table 2 and Fig. 1 summarise all models. All estimates are participant-level and out-of-sample.

Primary analysis. The pre-specified primary model (LR-L2 on all 288 features) did not separate the groups: AUC = 0.53 (95% CI 0.36–0.70), balanced accuracy = 0.54, permutation $p = 0.456$ (Fig. 2). The primary hypothesis, that task EEG features classify MDD above chance in unseen participants, is therefore not supported.

Demographic baseline. Age, sex and education alone gave AUC = 0.78 (0.63–0.91, $p = 0.002$). Of the three variables, only education differed between groups (Table 1), so it presumably carries most of this signal. This baseline outperformed the primary EEG model on the same folds (mean per-fold ΔAUC = -0.25, corrected resampled t-test $p = 0.019$; Table 3). It was not exceeded by any EEG model.

Secondary models. Of the secondary models, only L1-regularised logistic regression reached nominal significance: AUC = 0.72 (0.56–0.87), uncorrected permutation $p = 0.013$. This result does not survive Holm correction over the 13 secondary models with permutation tests ($p_{\text{Holm}} = 0.169$). Its advantage over the primary model on the same folds was also not significant (ΔAUC $= +0.16$, $p = 0.106$).

The nested model-selection estimate is the honest estimate of "choosing the best handcrafted-feature model", because the choice of model family is made inside the inner loop. It was lower: AUC = 0.63 (0.46–0.79, $p = 0.080$).

No single feature family was informative on its own. LR-L2 AUCs were:

- spectral and FAA: 0.51;
- ERP: 0.42;
- connectivity: 0.47;
- complexity: 0.55.

All permutation p-values were ≥ 0.325. Non-linear models (RBF SVM, random forest, LightGBM) did not improve on linear ones.

Trial-level models. Neither the Riemannian nor the deep-learning models exploited the trial-level data better than the handcrafted features (Table 2).

- Riemannian: tangent space + LR, AUC = 0.50; MDM, 0.35; ERP covariances, 0.61 ($p = 0.159$).
- Deep learning: EEGNet, AUC = 0.47; ShallowFBCSPNet, 0.44; TCN, 0.48. All confidence intervals include 0.5.

The deep networks reached high validation AUCs in some folds, but test AUCs fluctuated widely across folds. This is the signature of fitting participant-specific rather than diagnosis-specific structure with only 36 training participants per fold.

4.3 How much does trial-wise splitting inflate performance?

Fig. 3 contrasts the same two trial-level models under a trial-wise split, where trials of a participant can fall in both training and test sets, and a subject-wise split.

- **Tangent-space model.** Trial-level AUC was 0.86 with the trial-wise split and 0.51 with the subject-wise split. Averaging trial probabilities per participant gave a participant-level AUC of 1.00 under the trial-wise split, i.e. perfect apparent diagnosis, versus 0.50 under the subject-wise split.
- **Random forest on raw regional amplitudes**, the approach of many published studies and of an earlier unpublished analysis of these data by the author. Trial-level AUC was 0.77 versus 0.59, and participant-level AUC was 0.98 versus 0.70.

The trial-wise figures do not measure diagnosis. A classifier can recognise the *participant* from the individual statistics of their EEG, and the participant determines the label. Only the subject-wise figures answer the clinical question.

The subject-wise participant-level AUC of the raw-amplitude random forest (0.70) is the highest of any EEG model. We do not interpret it as evidence of a diagnostic signal, for three reasons:

- the model was included as a control, not pre-specified as a diagnostic model;
- it was evaluated with only three repetitions and no permutation test;

Table 2: Participant-level out-of-sample performance (MDD vs. HC, $n = 45$). Point estimates are means over 10 repetitions of 5-fold cross-validation (3 for deep learning). 95% CIs come from a participant bootstrap. Sensitivity and specificity are at a fixed 0.5 threshold. p_{perm}: label-permutation test re-running the full nested pipeline (number of permutations in parentheses); "–" means not run. The primary model is in bold.

Model	AUC [95% CI]	Bal. acc. [95% CI]	Sens.	Spec.	Brier	p_{perm}
Demographic baseline						
Age + sex + education (LR)	0.78 [0.63, 0.91]	0.76 [0.63, 0.88]	0.72	0.80	0.205	0.002 (1000)
Handcrafted features						
LR-L1, all features	0.72 [0.56, 0.87]	0.66 [0.51, 0.79]	0.67	0.64	0.223	0.013 (1000)
Nested model selection, all features	0.63 [0.46, 0.79]	0.60 [0.46, 0.74]	0.62	0.57	0.253	0.080 (200)
Random forest, all features	0.55 [0.38, 0.72]	0.55 [0.40, 0.70]	0.52	0.59	0.253	0.348 (200)
LightGBM, all features	0.55 [0.37, 0.72]	0.55 [0.41, 0.69]	0.40	0.70	0.325	0.353 (200)
LR-L2, all features (primary)	0.53 [0.36, 0.70]	0.54 [0.39, 0.69]	0.46	0.62	0.259	0.456 (1000)
Linear SVM, all features	0.51 [0.34, 0.68]	0.55 [0.40, 0.69]	0.65	0.45	0.246	0.479 (1000)
RBF SVM, all features	0.43 [0.27, 0.61]	0.50 [0.39, 0.62]	0.76	0.25	0.269	0.572 (200)
Feature families						
LR-L2, complexity	0.55 [0.38, 0.72]	0.57 [0.43, 0.72]	0.58	0.57	0.288	0.705 (1000)
LR-L2, spectral + FAA	0.51 [0.34, 0.68]	0.54 [0.40, 0.69]	0.48	0.60	0.282	0.325 (1000)
LR-L2, wPLI connectivity	0.47 [0.30, 0.65]	0.46 [0.32, 0.61]	0.39	0.54	0.304	0.768 (1000)
LR-L2, ERP	0.42 [0.25, 0.59]	0.43 [0.28, 0.57]	0.36	0.50	0.319	0.720 (1000)
Riemannian (trial level)						
ERP covariances + TS + LR	0.61 [0.43, 0.78]	0.56 [0.42, 0.70]	0.44	0.67	0.246	0.159 (200)
Tangent space + LR	0.50 [0.32, 0.67]	0.47 [0.32, 0.61]	0.39	0.54	0.278	0.532 (200)
MDM	0.35 [0.20, 0.52]	0.39 [0.25, 0.53]	0.28	0.50	0.290	0.891 (200)
Deep learning (trial level)						
TCN	0.48 [0.30, 0.65]	0.48 [0.34, 0.63]	0.41	0.56	0.287	–
EEGNet	0.47 [0.30, 0.65]	0.46 [0.35, 0.58]	0.17	0.75	0.324	–
ShallowFBCSPNet	0.44 [0.27, 0.61]	0.48 [0.35, 0.62]	0.31	0.65	0.350	–

- it is one of many analyses, and for reference, 5% of permutation runs of the primary pipeline exceeded AUC 0.68.

It does suggest that the raw ERP waveform may be worth testing as a pre-specified hypothesis in independent data.

4.4 Confounds and sensitivity analyses

Table 4 reports whether the EEG features could predict variables other than diagnosis, using the primary pipeline.

- **Sex:** AUC = 0.63 (0.45–0.79, $p = 0.018$).
- **Age above the median:** 0.50 ($p = 0.617$).
- **Recording period:** 0.55 ($p = 0.392$).

EEG features thus carried information about sex, consistent with known sex differences in EEG amplitude and spectra. Sex was balanced across groups (Table 1), however, so it cannot explain a diagnostic signal, and none was found. Neither age nor recording period was predictable from EEG.

Restricting the analysis to the 25 participants recorded between June and August 2015 (9 MDD), when both groups were recorded in parallel, gave AUC = 0.46 (0.24–0.70, $p = 0.517$).

Including all 53 participants, with no exclusions, gave AUC = 0.57 (0.42–0.72, $p = 0.278$). The null result is therefore not an artefact of the exclusion criteria. Primary-model AUC was similar in women (0.56, $n = 16$) and men (0.51, $n = 29$).

4.5 What, if anything, differs between groups?

Model-based importance. Permuting whole feature families in the test folds of the primary model changed AUC by:

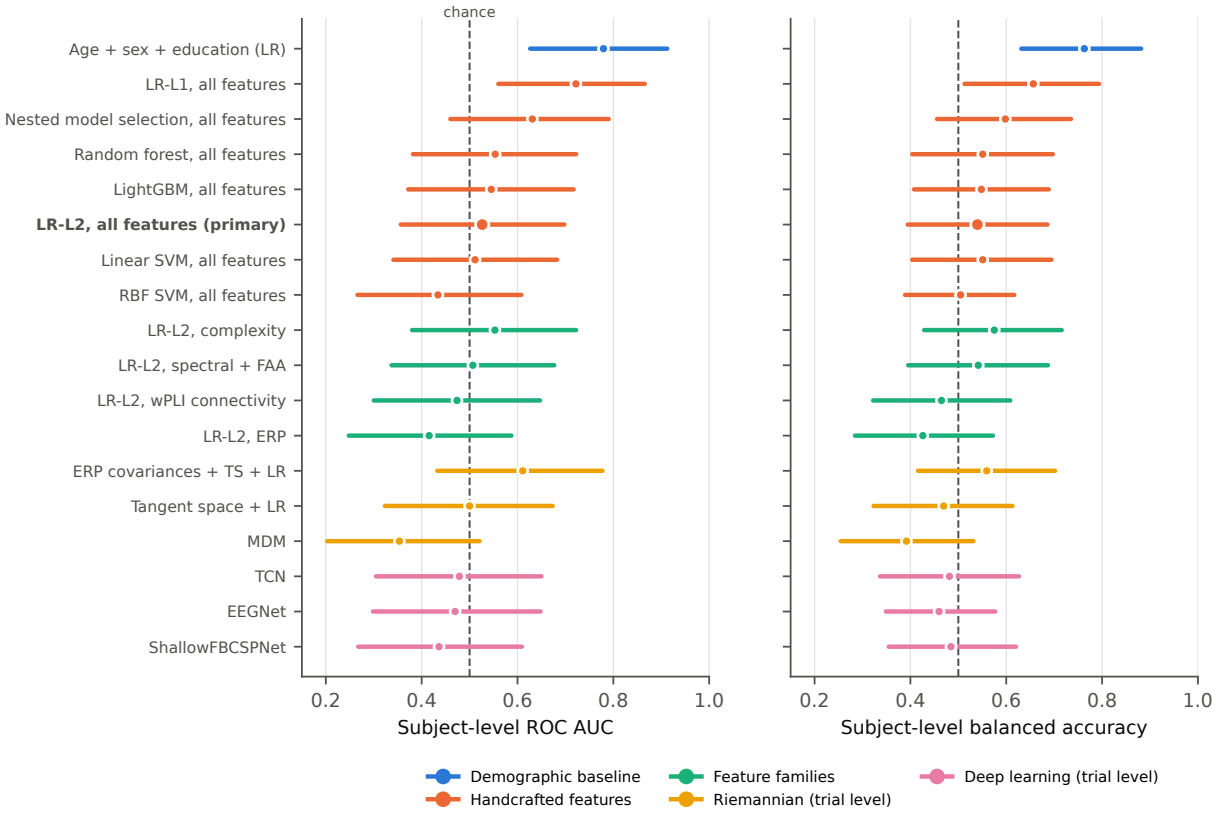


Figure 1: Participant-level ROC AUC (left) and balanced accuracy (right) with 95% bootstrap CIs, for every model. The dashed line marks chance. Only the demographic baseline (blue), which uses no EEG, has a CI clearly above chance.

- complexity: 0.083;
- spectral: −0.001;
- ERP: −0.004;
- connectivity: −0.030.

All bootstrap intervals included zero (Fig. 4a). SHAP values of the refitted primary model (Fig. 4b) were small in absolute terms, consistent with the strong regularisation selected.

Univariate effects. 17 of 288 features differed between groups at uncorrected $p < 0.05$, close to the ≈ 14 expected by chance. Only 1 survived false-discovery-rate correction: theta-band wPLI between left central and left temporal regions ($d = 1.38$, $p_{\mathrm{FDR}} = 0.036$; Table 5).

Spectral topography and ERPs. Fig. 5 suggests higher beta and low-gamma power in MDD, strongest at the edge of the montage, but no spatial cluster survived correction (minimum cluster $p = 0.06$).

Group-average ERPs (Fig. 6) showed the expected P1–N170 complex and a centro-parietal positivity in both groups. Visually, MDD participants had a smaller late positivity at occipito-temporal sites, but the spatio-temporal cluster test did not confirm any group difference (52 candidate clusters, minimum $p = 0.31$).

Table 3: Paired comparisons on identical participant folds. ΔAUC is the mean per-fold difference (model − reference). p_{corr} is from the corrected resampled t-test [42]. p_{Holm} applies Holm's correction to the permutation p-values of the secondary models.

	vs. primary		vs. demographics			
Model	ΔAUC	p_{corr}	ΔAUC	p_{corr}	p_{perm}	p_{Holm}
LR-L1, all features	+0.16	0.106	-0.09	0.470	0.013	0.169
Nested model selection, all features	+0.08	0.392	-0.17	0.176	0.080	0.955
Random forest, all features	+0.02	0.672	-0.22	0.029	0.348	1.000
LightGBM, all features	+0.01	0.944	-0.24	0.020	0.353	1.000
LR-L2, all features (primary)	–	–	-0.25	0.019	0.456	–
Linear SVM, all features	-0.13	0.422	-0.38	0.004	0.479	1.000
RBF SVM, all features	-0.13	0.429	-0.38	0.008	0.572	1.000
LR-L2, complexity	+0.06	0.517	-0.19	0.107	0.705	1.000
LR-L2, spectral + FAA	-0.01	0.903	-0.26	0.011	0.325	1.000
LR-L2, wPLI connectivity	-0.10	0.325	-0.34	0.008	0.768	1.000
LR-L2, ERP	-0.15	0.121	-0.40	0.000	0.720	1.000
ERP covariances + TS + LR	+0.08	0.455	-0.17	0.164	0.159	1.000
Tangent space + LR	-0.05	0.670	-0.30	0.027	0.532	1.000
MDM	-0.19	0.094	-0.44	0.000	0.891	1.000
TCN	-0.07	0.656	-0.31	0.078	–	–
EEGNet	-0.09	0.512	-0.33	0.022	–	–
ShallowFBCSPNet	-0.12	0.434	-0.36	0.025	–	–

Table 4: Can the EEG features predict confounds? The primary pipeline (LR-L2, all features) was applied to alternative targets, plus two sensitivity analyses of the MDD-vs-HC classification.

Target predicted from EEG features	n_{pos}/n	AUC [95% CI]	p_{perm}
sex (male)	29/45	0.63 [0.45, 0.79]	0.018
age > median	21/45	0.50 [0.33, 0.67]	0.617
recording period (>= Sep 2015)	18/45	0.55 [0.37, 0.72]	0.392
MDD vs. HC, primary pipeline, restricted subset			
Recorded Jun–Aug 2015	9/25	0.46 [0.24, 0.70]	0.517
All 53 participants (no exclusions)	24/53	0.57 [0.42, 0.72]	0.278

5 Discussion

5.1 Main finding

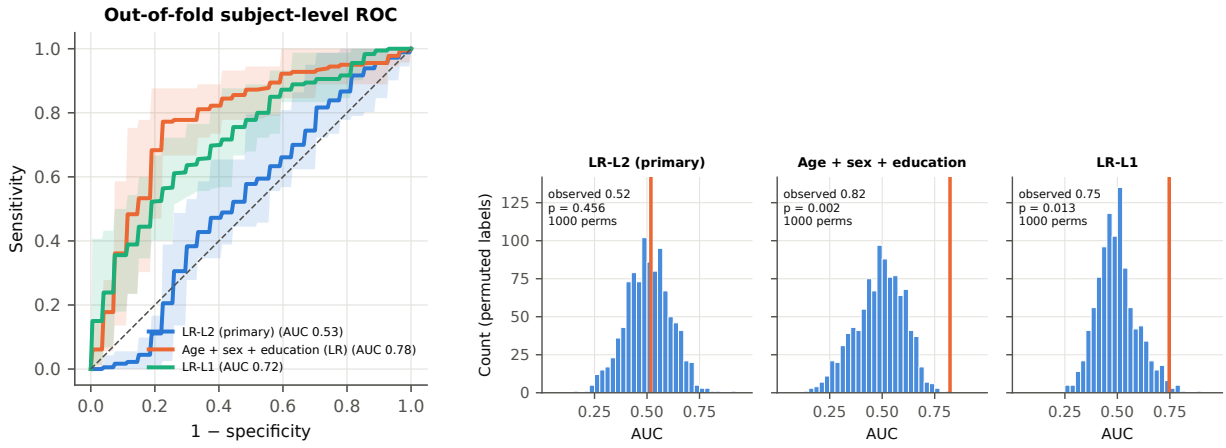
Under a strictly participant-wise, pre-specified protocol, task EEG from the MODMA dot-probe recordings did not separate MDD from healthy controls better than chance. The conclusion holds across all model types tested:

- the pre-specified primary model and every feature family on its own;
- linear, kernel and tree-based classifiers;
- Riemannian models and three deep networks trained on the full 128-channel epochs.

The one nominally significant secondary result (L1-regularised logistic regression) did not survive correction for the number of secondary models. The honest estimate of picking the best model, obtained by nested model selection, fell to 0.63, with a confidence interval including chance.

By contrast, three demographic variables (age, sex and education) reached AUC 0.78. Any EEG model that claims diagnostic value on this dataset should be benchmarked against this figure, not only against 0.5.

These findings agree with reviews describing EEG markers of depression as small and heterogeneous effects [8–10]. They also agree with the meta-analytic null result for frontal alpha asymmetry [12]. They do not agree with the near-perfect accuracies often reported for EEG-based depression detection [17].



(a) **(b)**

Figure 2: (a) Mean out-of-fold ROC curves across repetitions (band: 2.5–97.5% range over repetitions). (b) Permutation null distributions of the participant-level AUC, with the observed AUC (first repetition) marked.

5.2 Why do published accuracies differ so much?

Our inflation control offers a direct explanation. On the same data, the tangent-space model evaluated with a trial-wise split ranked every participant correctly (participant-level AUC 1.00; trial-level 0.86). Evaluated on held-out participants, it was at chance (0.50).

When epochs of one person appear in both training and test sets, the classifier only needs to recognise the person, and every person carries one label. Individual EEG signatures are highly distinctive, so this task is easy, and it says nothing about diagnosing a new patient [1, 3]. Reported accuracies therefore cannot be interpreted unless the split is stated to be across participants. In this light, values above 95% on datasets of a few dozen participants should be regarded with suspicion.

5.3 Model selection and multiplicity

With $n = 45$, sampling noise alone produces large AUCs: under label permutation, the primary pipeline exceeded AUC 0.68 in 5% of runs. Across 18 configurations, some will exceed that by chance. Reporting the best of them, here L1-LR at AUC 0.72, would have yielded a publishable but misleading result.

Three elements of the design guard against this:

- a primary model and outcome fixed before any result existed;
- multiplicity correction across secondary models;
- a nested estimate of the selection procedure itself [19].

We recommend all three as standard practice for small clinical EEG datasets.

5.4 Confounds

The groups differed in education and, partly, in recording period: some months contain only patients or only controls. In principle, period effects (equipment state, staff, season) could produce group differences unrelated to depression.

Two observations make this unlikely to be the explanation here. First, the EEG features did not predict recording period above chance. Second, the analysis restricted to the months in which both groups were recorded was no more successful than the full analysis.

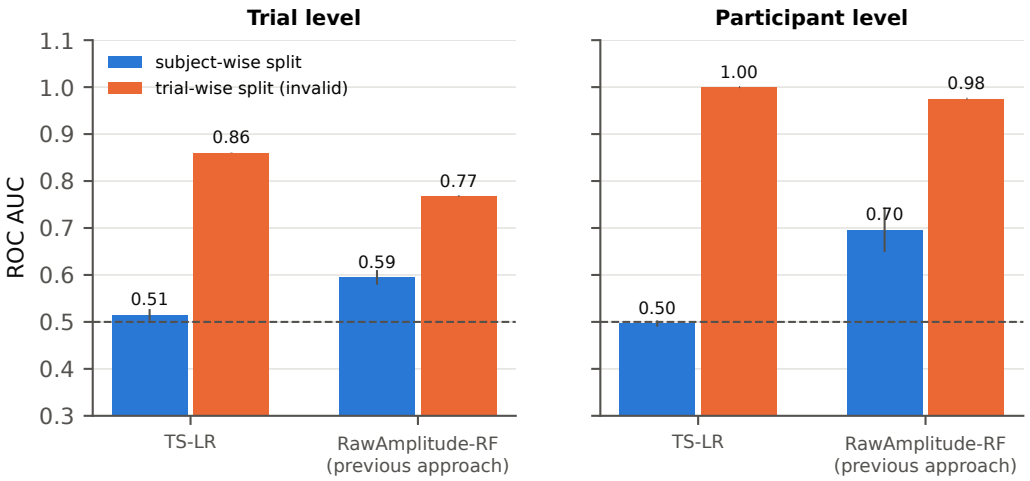


Figure 3: AUC of the same models evaluated with a trial-wise split (invalid; trials of each participant appear in training and test) and with a subject-wise split, at the trial level (left) and after averaging trial probabilities per participant (right). Bars show means over three repetitions, with SD as error bars. The trial-wise split is shown only to quantify the inflation.

The signal-quality measures did not differ between groups. Even so, one descriptive pattern deserves caution:

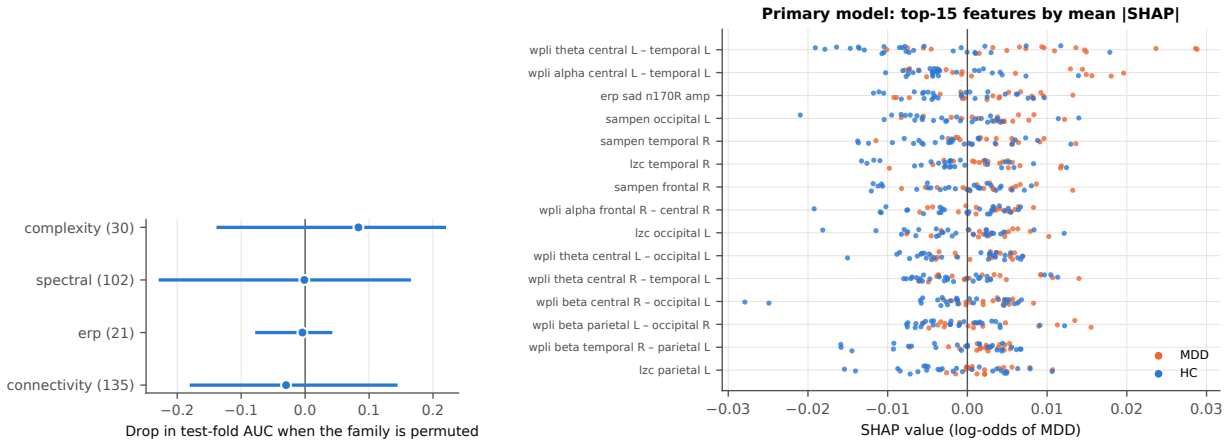
- MDD participants showed numerically higher beta and low-gamma power, strongest at peripheral electrodes;
- they also showed higher entropy and Lempel–Ziv complexity, the family whose permutation importance was largest, though not significant.

Both patterns are also typical signatures of residual muscle activity. ICA with ICLabel removes the clearest muscle components but not all EMG. Any future report of high-frequency or complexity differences in these data should therefore rule out muscle tension before attributing them to depression.

The single connectivity feature that survived FDR correction (left central–temporal theta wPLI) was not robustly used by any model. It should be considered a hypothesis for replication, not a finding.

5.5 Limitations

- **Sample size.** With 18 MDD and 27 HC participants after exclusion, the participant-level confidence intervals are wide: for the primary model, 0.36–0.70. A modest true effect, around AUC 0.6, cannot be excluded. Our conclusion is that these data provide no evidence of a diagnostic signal, not that none exists.
- **Single site and single task.** All data come from one site, one EEG system and one paradigm. Clinical details that may modulate EEG are not modelled, notably medication status and illness duration.
- **Asymmetric exclusions.** The pre-specified quality criteria excluded more patients than controls. The analysis on all 53 participants, however, gave the same conclusion.
- **Deep learning budget.** For computational reasons, the deep networks were evaluated with 3 rather than 10 repetitions and without permutation tests. Their confidence intervals all included chance, and their point estimates were below that of the primary model.



(a) **(b)**

Figure 4: (a) Drop in test-fold AUC of the primary model when each feature family is permuted jointly (mean and 95% interval over 50 outer folds; number of features in parentheses). (b) SHAP values of the primary model refitted on all participants, for the 15 features with the largest mean |SHAP| (descriptive).

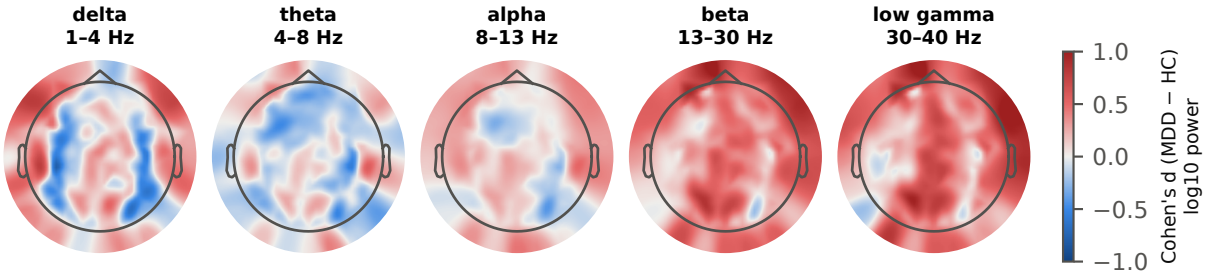


Figure 5: Group differences in log band power per electrode (Cohen's d, MDD − HC). Electrodes belonging to a significant cluster (cluster permutation test, $p < 0.05$) would be marked; none were.

- **Feature space.** The features were fixed a priori from the literature. Other representations could perform better, but searching for them on the same 45 participants would reintroduce the selection bias this study was designed to avoid.

5.6 Conclusion

Evaluated rigorously on participants never seen by the model, 128-channel task EEG from the MODMA dot-probe dataset did not distinguish major depression from healthy controls, and was outperformed by a demographic baseline. The large gap between trial-wise and participant-wise estimates on the same data shows how evaluation choices can manufacture apparent diagnostic accuracy. Credible progress towards EEG markers of depression will require participant-wise evaluation, pre-specified analyses, demographic baselines and substantially larger, multi-site samples.

Code and data availability

All code, the analysis plan (fixed and version-controlled before any result was produced) and the list of deviations are available in the accompanying repository. The EEG data belong to the MODMA project [6] and were obtained from its maintainers for academic use; they are not redistributed.

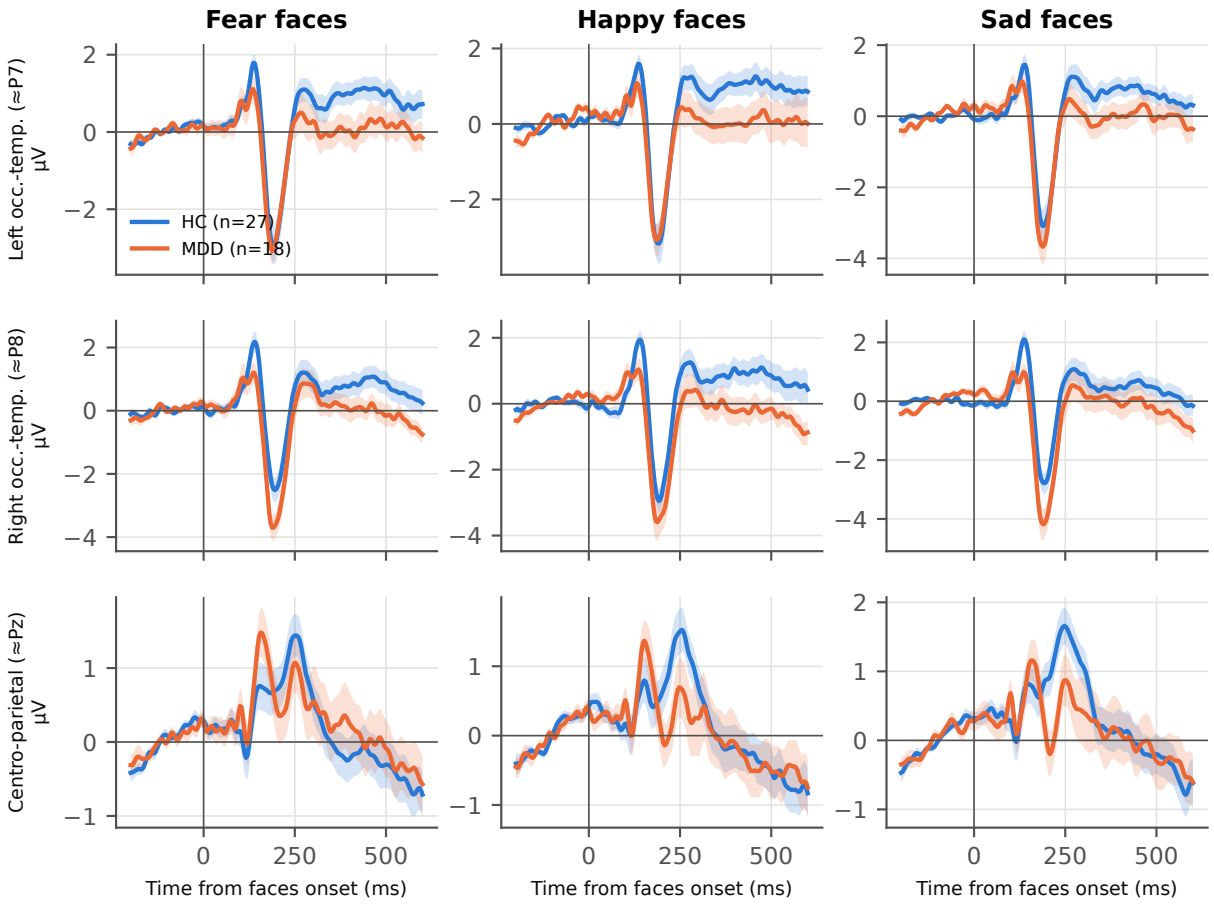


Figure 6: Grand-average ERPs (±SEM) per group and face condition, at the left and right occipito-temporal clusters and the centro-parietal cluster. Shaded windows are the a-priori N170 (140–200 ms) and P300 (250–500 ms) measurement windows.

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Use of AI tools

An AI coding assistant was used to write and test the analysis code and to draft the text. All analysis decisions, code and results were checked by the author, and every number in this paper is generated programmatically from saved outputs.

Disclaimer

This work is a methodological study. The models described here are *not* diagnostic tools and must not be used for clinical decision-making.

References

[1] Sohrab Saeb, Luca Lonini, Arun Jayaraman, David C. Mohr, and Konrad P. Kording. The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5):1–9, 2017.

[2] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

[3] Geoffrey Brookshire, Jake Kasper, Nicholas M. Blauch, Yunan Charles Wu, Ryan Glatt, David A. Merrill, Spencer Gerrol, Keith J. Yoder, Colin Quirk, and Ché Lucero. Data leakage in deep learning studies of translational EEG. *Frontiers in Neuroscience*, 18:1373515, 2024. doi: 10.3389/fnins.2024.1373515.

[4] Gaël Varoquaux. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage*, 180:68–77, 2018.

[5] Etienne Combrisson and Karim Jerbi. Exceeding chance level by chance: The caveat of theoretical chance levels in brain signal classification and statistical assessment of decoding accuracy. *Journal of Neuroscience Methods*, 250:126–136, 2015.

[6] Hanshu Cai et al. A multi-modal open dataset for mental-disorder analysis. *Scientific Data*, 9:178, 2022.

[7] Colin MacLeod, Andrew Mathews, and Philip Tata. Attentional bias in emotional disorders. *Journal of Abnormal Psychology*, 95(1):15–20, 1986.

[8] Sebastian Olbrich and Martijn Arns. EEG biomarkers in major depressive disorder: Discriminative power and prediction of treatment response. *International Review of Psychiatry*, 25 (5):604–618, 2013. doi: 10.3109/09540261.2013.816269.

[9] Fernando Soares de Aguiar Neto and João Luís Garcia Rosa. Depression biomarkers using non-invasive EEG: A review. *Neuroscience & Biobehavioral Reviews*, 105:83–93, 2019. doi: 10.1016/j.neubiorev.2019.07.021.

[10] Leif Simmatis, Emma E. Russo, Joseph Geraci, Irene E. Harmsen, and Nardin Samuel. Technical and clinical considerations for electroencephalography-based biomarkers for major depressive disorder. *npj Mental Health Research*, 2:18, 2023. doi: 10.1038/s44184-023-00038-7.

[11] John J. B. Allen and Samantha J. Reznik. Frontal EEG asymmetry as a promising marker of depression vulnerability: summary and methodological considerations. *Current Opinion in Psychology*, 4:93–97, 2015.

[12] Nikita van der Vinne, Madelon A. Vollebregt, Michel J. A. M. van Putten, and Martijn Arns. Frontal alpha asymmetry as a diagnostic marker in depression: Fact or fiction? A meta-analysis. *NeuroImage: Clinical*, 16:79–87, 2017.

[13] Shlomo Bentin, Truett Allison, Aina Puce, Erik Perez, and Gregory McCarthy. Electrophysiological studies of face perception in humans. *Journal of Cognitive Neuroscience*, 8(6):551–565, 1996.

[14] John Polich. Updating P300: an integrative theory of P3a and P3b. *Clinical Neurophysiology*, 118(10):2128–2148, 2007.

[15] Greg Hajcak, Annmarie MacNamara, and Doreen M. Olvet. Event-related potentials, emotion, and emotion regulation: an integrative review. *Developmental Neuropsychology*, 35(2):129–155, 2010.

[16] Xiaowei Li, Jianxiu Li, Bin Hu, et al. Attentional bias in MDD: ERP components analysis and classification using a dot-probe task. *Computer Methods and Programs in Biomedicine*, 164:169–179, 2018. doi: 10.1016/j.cmpb.2018.07.003.

[17] Betul Ay, Ozal Yildirim, Muhammed Talo, Ulas Baran Baloglu, Galip Aydin, Subha D. Puthankattil, and U. Rajendra Acharya. Automated depression detection using deep representation and sequence learning with EEG signals. *Journal of Medical Systems*, 43(7):205, 2019. doi: 10.1007/s10916-019-1345-y.

[18] ZhongYi Zhang et al. Cross-subject depression level classification using EEG signals with a sample confidence method. *arXiv preprint arXiv:2503.13475*, 2025.

[19] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[20] Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti Hämäläinen. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*, 7:267, 2013.

[21] Nima Bigdely-Shamlo, Tim Mullen, Christian Kothe, Kyung-Min Su, and Kay A. Robbins. The PREP pipeline: standardized preprocessing for large-scale EEG analysis. *Frontiers in Neuroinformatics*, 9:16, 2015.

[22] Pierre Ablin, Jean-François Cardoso, and Alexandre Gramfort. Faster independent component analysis by preconditioning with Hessian approximations. *IEEE Transactions on Signal Processing*, 66(15):4040–4049, 2018.

[23] Luca Pion-Tonachini, Ken Kreutz-Delgado, and Scott Makeig. ICLabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage*, 198:181–197, 2019.

[24] Adam Li et al. MNE-ICALabel: Automatically annotating ICA components with ICLabel in Python. *Journal of Open Source Software*, 7(76):4484, 2022.

[25] Martin Vinck, Robert Oostenveld, Marijn van Wingerden, Franscesco Battaglia, and Cyriel M. A. Pennartz. An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*, 55(4):1548–1565, 2011.

[26] Joshua S. Richman and J. Randall Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology-Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000.

[27] Tomoyuki Higuchi. Approach to an irregular time series on the basis of the fractal theory. *Physica D*, 31(2):277–283, 1988.

[28] Abraham Lempel and Jacob Ziv. On the complexity of finite sequences. *IEEE Transactions on Information Theory*, 22(1):75–81, 1976.

[29] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[30] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.

[31] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[32] Alexandre Barachant, Stéphane Bonnet, Marco Congedo, and Christian Jutten. Multiclass brain–computer interface classification by Riemannian geometry. *IEEE Transactions on Biomedical Engineering*, 59(4):920–928, 2012.

[33] Marco Congedo, Alexandre Barachant, and Rajendra Bhatia. Riemannian geometry for EEG-based brain-computer interfaces; a primer and a review. *Brain-Computer Interfaces*, 4(3): 155–174, 2017.

[34] Yilun Chen, Ami Wiesel, Yonina C. Eldar, and Alfred O. Hero. Shrinkage algorithms for MMSE covariance estimation. *IEEE Transactions on Signal Processing*, 58(10):5016–5029, 2010.

[35] Vernon J. Lawhern, Amelia J. Solon, Nicholas R. Waytowich, Stephen M. Gordon, Chou P. Hung, and Brent J. Lance. EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5):056013, 2018.

[36] Robin Tibor Schirrmeister, Jost Tobias Springenberg, Lukas Dominique Josef Fiederer, Martin Glasstetter, Katharina Eggensperger, Michael Tangermann, Frank Hutter, Wolfram Burgard, and Tonio Ball. Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*, 38(11):5391–5420, 2017.

[37] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 2018.

[38] Lukas A. W. Gemein, Robin T. Schirrmeister, Patryk Chrabąszcz, Daniel Wilson, Joschka Boedecker, Andreas Schulze-Bonhage, Frank Hutter, and Tonio Ball. Machine-learning-based diagnostics of EEG pathology. *NeuroImage*, 220:117021, 2020.

[39] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[40] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57 (1):289–300, 1995.

[41] Eric Maris and Robert Oostenveld. Nonparametric statistical testing of EEG- and MEG-data. *Journal of Neuroscience Methods*, 164(1):177–190, 2007.

[42] Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003.

A Deviations from the pre-specified plan

Deviations D1–D4 were decided before any model had been trained; at that point only the preprocessing log had been inspected. D5 was decided after the subject-level models had been evaluated but before any result of the affected (Riemannian) models existed.

D1 **Bad-channel detector.**

- *Problem:* the pre-specified rule (flat signal, robust log-SD $z > 5$, median neighbour correlation < 0.4) flagged up to 48 of 129 channels in a recording. Of all flags, 94% came from the neighbour-correlation criterion, and they included channels adjacent to the vertex reference with *negative* neighbour correlations. Whole-recording correlation is not a valid quality measure on reference-dominated, artefact-laden data.
- *Change:* the rule was replaced by the PREP implementation [21].
- *Unchanged:* the 20% exclusion threshold.

D2 **Re-run.** Fourteen participants were re-run after an infrastructure failure: a library was being installed during the first run. Since D1 required a full re-run anyway, all 53 were re-processed.

D3 **Added model.** The ERP-covariance Riemannian model was added as a secondary model.

D4 **Added sensitivity analysis.** Exclusions affected 6 MDD and 2 HC participants, so we added a sensitivity analysis of the primary model on all 53 participants.

D5 **Computational budget of the Riemannian models.**

- *Problem:* timing one outer fold showed that the full protocol (10 repetitions plus 200 permutations, i.e. 1,050 outer folds per model) would take more than 12 h, mostly because per-trial covariances were recomputed in every inner-loop fit.
- *Change 1 (no effect on results):* for the tangent-space and MDM models, per-trial OAS covariances were computed once. Each covariance depends only on its own trial, so this involves no learning across samples; the tangent-space reference point, scaling, classifier and MDM centroids were still fitted inside each training fold. A unit test verifies identical predictions.
- *Change 2:* the ERP-covariance model, whose class prototypes depend on labels and must be recomputed within each fold, used a fixed regularisation $C = 0.1$ instead of an inner grid. A value fixed in advance cannot inflate performance; at worst it is slightly sub-optimal.

B Preprocessing quality

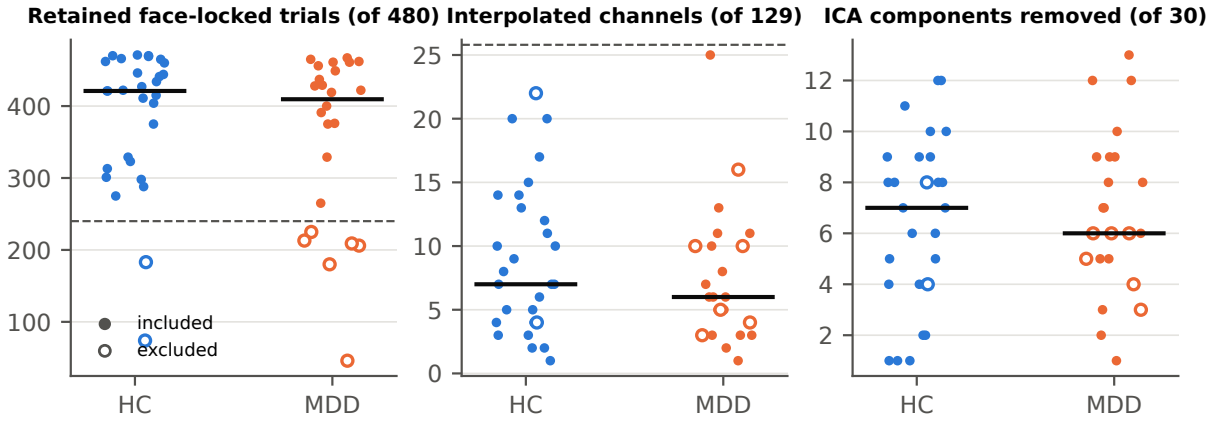


Figure 7: Signal quality per participant: retained face-locked trials, interpolated channels and removed ICA components. Open markers show participants excluded by the pre-specified criteria (dashed lines). Horizontal bars are group medians.

C Univariate group differences

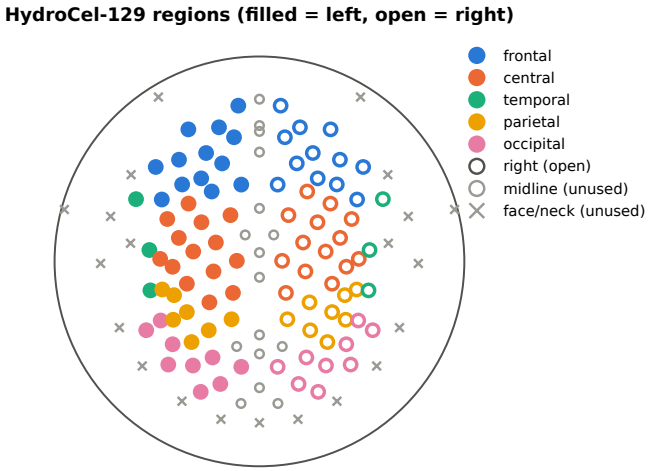


Figure 8: Assignment of HydroCel-129 electrodes to the ten regions, obtained by nearest 10–10 anchor on the fitted head sphere. Filled markers are left-hemisphere electrodes and open markers right-hemisphere electrodes. Grey electrodes (midline, and the face/neck ring more than 115° from the vertex) are not used for regional features.

Table 5: The ten features with the smallest uncorrected Mann–Whitney p (MDD vs. HC, included participants). p_{FDR} is Benjamini–Hochberg corrected over all 288 features.

Feature	Family	Cohen's d	p	p_{FDR}
`wpli_theta_central_L__temporal_L`	connectivity	1.38	<0.001	0.04
`sampen_occipital_L`	complexity	0.79	0.003	0.45
`rel_lowgamma_occipital_L`	spectral	0.57	0.005	0.47
`sampen_temporal_R`	complexity	0.76	0.009	0.62
`rel_beta_occipital_L`	spectral	0.69	0.012	0.66
`wpli_beta_central_R__occipital_L`	connectivity	-0.70	0.018	0.66
`sampen_parietal_L`	complexity	0.72	0.020	0.66
`lzc_occipital_L`	complexity	0.71	0.022	0.66
`lzc_temporal_R`	complexity	0.72	0.022	0.66
`pow_lowgamma_occipital_L`	spectral	0.64	0.024	0.66